Cyber-bullying as an impediment in the dynamic morality algorithm development

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1. Abstract

This paper uses our cyber-bullying research conducted on Polish students to presents the problems of cyber-bullying for Machine Ethics research. Focused on both Internet culture and hate phenomena, we present the challenges for a dynamic moral algorithm including but not limited to: irony and steganography of picture-text context. Final piece of this paper presents ideas for future work in overcoming highlighted inpediments.

2. Preface

Technology's presence in our lives and its influence on it makes us not only reconsider its place but also – ours. For centuries it was the technological revolutions that brought social changes. Whether it was inventing agriculture which allowed us to change nomadic lifestyle to settled (which brought grow of population) or replace manufactures with factories in the industrial revolution that moved people from country side to cities.

Today technology does not only change the society but makes us reconsider our roles in it, e.g. children are getting more and more tech-savvy on one hand but on the other – growing number of older people can not find their feet in the technological world and they are being guided by the younger generation.

This tendency challenges modern teachers who have to not only stay up to date with their subject but also improve ways to communicate with pupils, including technological and cultural contexts in which they find themselves everyday. This growing and changing rapidly fast Internet trends alone require an enormous time commitment to

3. The preconfigurative society

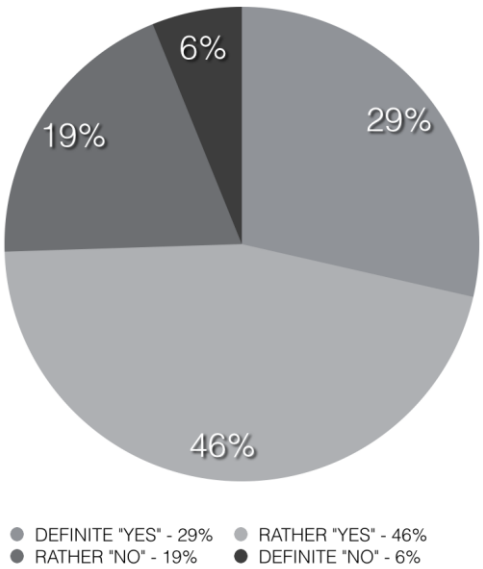
For ages it had been the elders that led the community with their experience and wisdom. Multigenerational, patriarchal households were organized by the criteria of age and sex. Today more often children are becoming teachers for their parents and grandparents. This phenomena is known as the “preconfigurative society” and verifying its scale is one of the main goals of our research, introduced in this papers.

4. The questionnaire

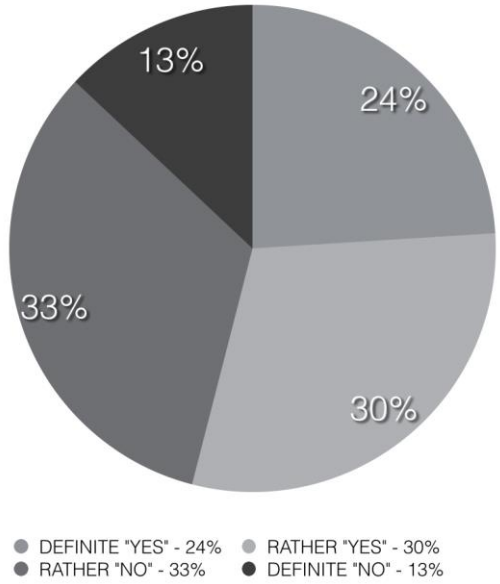
To research young people's approach to using the Internet, we have selected a group of 100 students: 44 boys and 56 girls with around 70% of an average age between 13-15 years old and 30% between 16-18 years old.

We have prepared 18 sentences in which respondents were asked to determine, how much they agree with presented statements on a 4-grade scale: “definite 'Yes'”, “rather 'Yes'”, “rather 'No'” and “definite 'No'”.

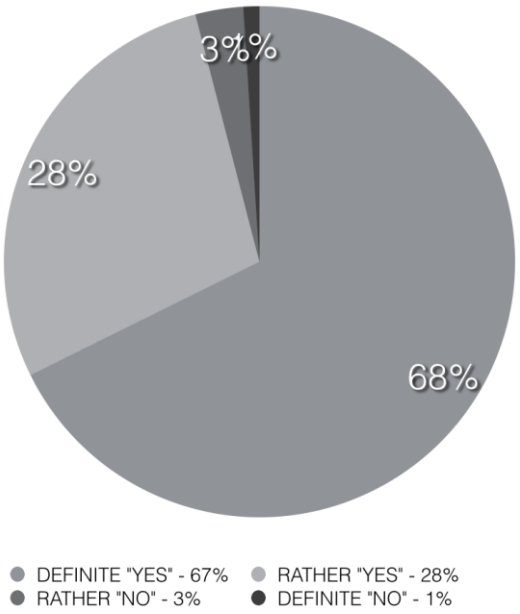
Sentences 1-4 were focused on being tech-savvy and included the preconfigurative society phenomena verification. Our conclusions:



1. Every third respondent is sure and almost half of them suspects that they could diagnose and solve a problem with their computer.



2. Over a half of respondents is at least “rather asked” to fix or help with a computer problem.

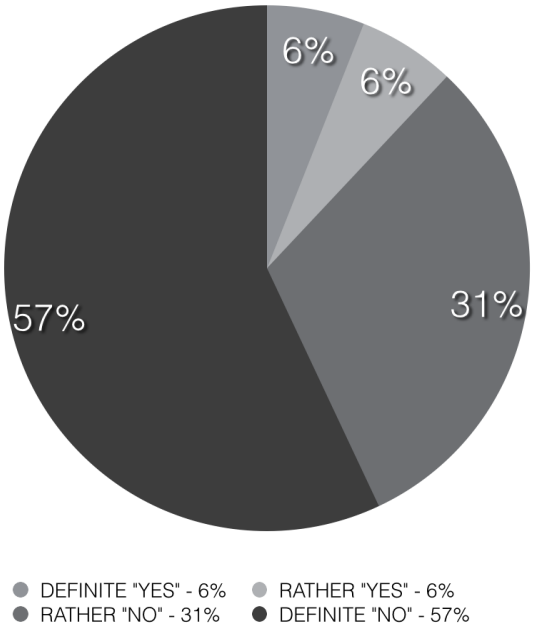


3. Only 1 in 100 respondents has a definite problem with using new devices.

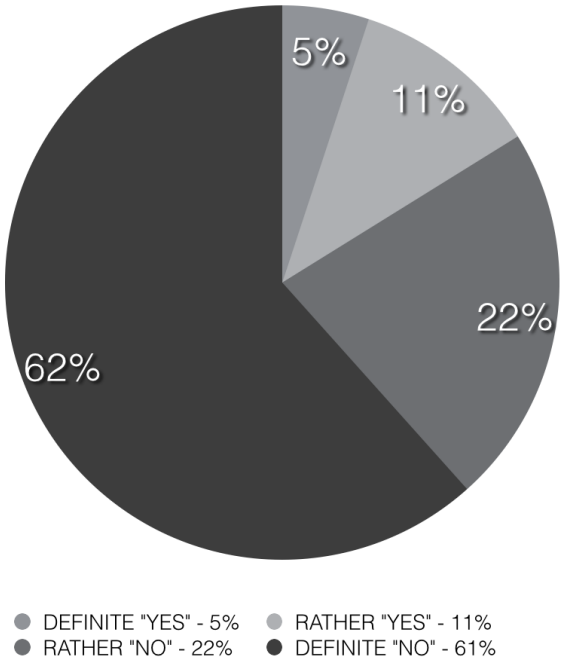
Secondly we asked about the way respondents use the Internet:

1. Over 90% of respondents use the Internet for school and homework.
2. Less than half of respondents uses the Internet to promote their artistic work.

In questions on cyber-bullying, almost 9 out of 10 people denied posting negative comments on the Web:



Also, 2 out of 3 respondents is affraid to do wrong things on the Internet:



On the other hand, 40% of surveyed people admit to browse other people’s photos in secret but every third respondent definitely denies doing so.

Sentences 13-16 were focused on yet another popular trend, the Fear and Joy of Missing Out. Over 50% are sure and 20% are likely to stop studying and check the notifications on their devices what in combination with losing track of time during web and SNS browsing might be a major threat for their education and social life. Fortunately, our survey showed that 83% of respondents likes “unplugged” entertainment and over a half of respondents even declares to “switch off” from time to time.

When it came to privacy questions, the respondents gave most even answers with exactly 50% having at least some problems with sharing their true personal data on the Web and the other half not finding it difficult. Though, only 31% of them is sure that it is possible to be fully anonymous on-line.

5. The challenges

Our questionnaire gave us a huge insight into the matter on how people use the Internet and what is their approach in using technology at all.

Our previously introduced Artificial Moral Agent uses an emotion-extraction algorithm to base its moral judgments on and the results of the survey has ensured us that our web-crawler should differ bullying expressions from negative emotions and exclude them.

Though containing same kind of emotions, hateful comments present doing wrong as good. Since Aristotle we have defined “good” as “what everybody desires” (lat. *bonum est, quod omnes appetunt*) and this statistical, common-sense truth finds it way also in Machine Ethics research. This could end up in finding ”killing somebody” a “joyful” or “pleasant” experience. This problem was also briefly addressed during out last LAU presentation.

Secondly, especially when working with young people, one has to consider the phenomena of steganography. Unlike cryptography, which is focused on hiding the message itself, steganography puts it in plain sight – usually with a graphic context – and only while understanding the correlation, one can understand the true communique. on-line.

6. The solutions

Our In our survey, we had asked the respondents if they have ever posted a hateful comment about any celebrity, famous athlete or a politician and got a positive feedback from 42% of the respondents. In the next question we made case more specific and asked if they have ever posted a negative comment about any given person:

1. Howard Webb; football referee who got cyber-bullied in Poland for awarding Austrian team a penalty kick in the UEFA Euro 2008 games,
2. Justin Bieber; pop singer whose music video is one of the most disliked clips on YouTube,
3. Donald Tusk, Polish Prime Minister,
4. Pope Francis.

Howard Webb	39%
Justin Bieber	64%
Donald Tusk	52%
Pope Francis	20%

With such promising correlation between famous people and hateful comments on the Web, the web-crawling agent should be possible to separate negative emotions from hateful comments, classifying results from texts including celebrities as the latter. By that, newly created data-base will become more dependable and be used for risk-free Machine Ethics research.

Another way to improve the moral algorithm will be implementing a meme based phrase library that will allow to extract right emotions from Internet-based speech which gets more and more often moved from graphic to text message.

7. Future work

Due to its promising contribution to our Machine Ethics research, we will expand our survey on higher number and more diverse respondents. This should eventually lead to understand the web-crowd better and improve our Agent's moral judgement.

Secondly, we will create a separate date-base with suspected cyber-bullying expressions and work more closely on contextual possibilities of differing them from regular negative emotion expressions.

Building an Internet-meme phrasebook with emotion correlation might be a useful tool not only for AI researchers but also an indispensable resource for teachers who.

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